

CYG Wayon 800V WMOS™ M3 Power MOSFET

The new products released from CYG Wayon advanced super junction technology

Introduction

CYG Wayon 800V WMOS™ M3 is a new silicon-based cutting-edge technology, which features extremely low gate charge for high efficiency when switching, especially for high frequency switching. With enhanced design, the new WMOS™ M3 family has also strong surge performance for LED application.

How does Power MOS affect efficiency and temperature?

Switching power loss is related to MOSFET Qg, lower Qg lead to lower MOS switching power loss. Conduction power loss is related to MOSFET Rdson, lower Rdson lead to lower MOS conduction power loss.

FOM(Rdson*Qg) is a useful method to compare power MOSFET. Lower FOM means higher efficiency and lower MOS temperature. CYG Wayon 800V WMOS™ M3 series shows best FOM, better efficiency and temperature.

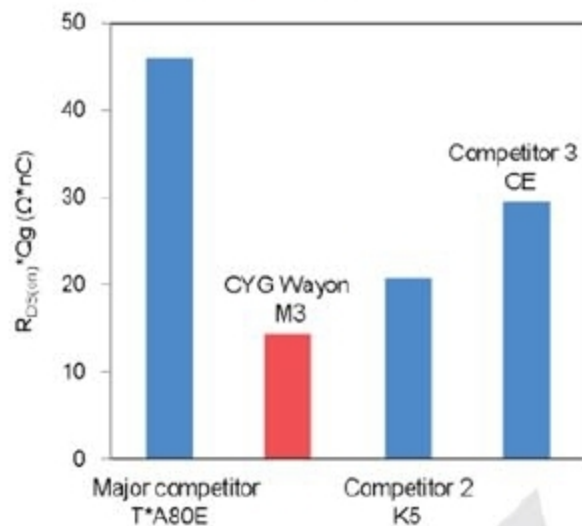


Figure 1. 800V Power MOS FOM comparison

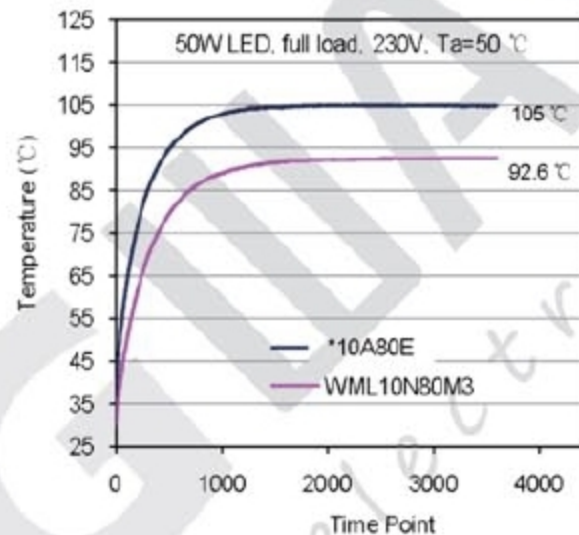


Figure 3. Temperature Comparison

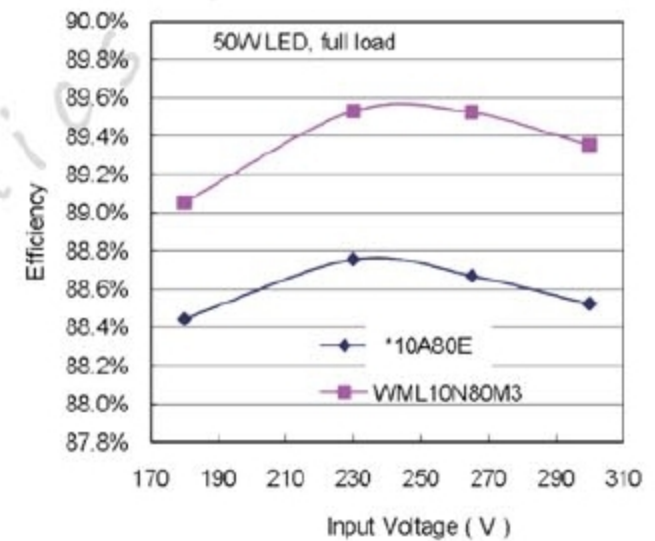


Figure 2. Efficiency Comparison

Main Applications

●Charger/ Adapter ●PC ●TV ●LED /Lighting ●UPS /Server

$R_{DS(on), max} [\Omega]$	TO-252	TO-251	TO-220F	TO-220	TO-263	TO-262	TO-247
4.0	WMO03N80M3	WMP03N80M3	WML03N80M3	WMK03N80M3	WMM03N80M3	WMN03N80M3	
2.3	WMO05N80M3	WMP05N80M3	WML05N80M3	WMK05N80M3	WMM05N80M3	WMN05N80M3	
2.0	WMO06N80M3	WMP06N80M3	WML06N80M3	WMK06N80M3	WMM06N80M3	WMN06N80M3	
1.8	WMO07N80M3	WMP07N80M3	WML07N80M3	WMK07N80M3	WMM07N80M3	WMN07N80M3	
1.38	WMO08N80M3	WMP08N80M3	WML08N80M3	WMK08N80M3	WMM08N80M3	WMN08N80M3	
1.03	WMO10N80M3	WMP10N80M3	WML10N80M3	WMK10N80M3	WMM10N80M3	WMN10N80M3	
0.8	WMO11N80M3	WMP11N80M3	WML11N80M3	WMK11N80M3	WMM11N80M3	WMN11N80M3	
0.62	WMO12N80M3	WMP12N80M3	WML12N80M3	WMK12N80M3	WMM12N80M3	WMN12N80M3	WMJ12N80M3
0.48	WMO13N80M3	WMP13N80M3	WML13N80M3	WMK13N80M3	WMM13N80M3	WMN13N80M3	WMJ13N80M3
0.36			WML15N80M3	WMK15N80M3	WMM15N80M3	WMN15N80M3	WMJ15N80M3
0.26			WML25N80M3	WMK25N80M3	WMM25N80M3	WMN25N80M3	WMJ25N80M3
0.195			WML30N80M3	WMK30N80M3	WMM30N80M3	WMN30N80M3	WMJ30N80M3
0.128			WML35N80M3				WMJ35N80M3